

MTRN4231: Project README

1. Due Dates and Weighting

	Due Date	Weighting
README [^]	17:00 Friday Week 12	20%
Peer Evaluation (see Section 3.2)	17:00 Friday Week 12	-

[^]Note that the README mark will be scaled by the results of the peer assessment activity – see Section 3.2.

2. Purpose

This assessment gives your team the opportunity to document your robotic system in a professional, industry-standard format.

You will prepare a GitHub README file that clearly communicates the purpose, design, architecture, and usage of your system. The README should enable someone new to the project (such as a potential employer, research collaborator, or other student) to understand what you built, how it works, and how to reproduce it.

This assessment focuses on technical communication and system understanding. It will demonstrate your team's ability to:

- Present a clear and professional overview of your project and its purpose.
- Document a complex ROS2 system and explain its architecture.
- Provide sufficient information for someone else to install, run and evaluate your system.
- Summarise the design and innovation that make your solution unique.

3. Task Description

You will collaboratively prepare a structured GitHub README file for your MTRN4231 project repository. It must be written in Markdown¹ (README.md), use professional language and include visual and structural elements such as tables, diagrams and images where appropriate.

Your README must include the required sections given in Section 3.1.1 and will be marked according to the rubric in Section 4.

In addition, you will participate in a peer assessment activity on Moodle to evaluate your team members' contributions and teamwork

3.1. README

This README serves three key purposes. It allows you, as a team, to:

1. Showcase your robotic system and how it fulfils the customer requirements.
2. Clearly document your ROS2 architecture, operation and integration of all required components.
3. Communicate your design process, technical achievements and opportunities for improvement in a concise, professional way.

¹ <https://daringfireball.net/projects/markdown/> <https://www.markdownguide.org/>

The README should be aimed at a technically literate reader, such as another robotics student or engineer aiming to build on, or reproduce, your team's work. Avoid unnecessary filler text – focus on clarity, completeness and professionalism.

As a guide, it is expected that the README be around 1,500-2,500 words (around 4-6 pages when rendered into PDF). This is a *guide*, however, *not a limit or a target* – focus on clarity, conciseness and meeting the specifications in this document rather than length.

3.1.1. Requirements

The README must have the following sections:

1. Table of Contents:

- Include a Table of Contents at the start of your README (this can be auto-generated).

2. Project Overview:

- A short description of the task or problem your system solves, including the intended “customer” or end-user.
- A summary of the robot's functionality.
- A short video (e.g. 10-30s) of the robot completing one full cycle/operation demonstrating closed-loop behaviour and visualisation. (This could be embedded or an external link to a YouTube/OneDrive/Google Drive video.)

3. System Architecture:

- A diagram of your ROS2 nodes, topics, services and actions (e.g. from `rqt_graph` or a custom schematic).
- A package-level architecture diagram showing node interactions and topics.
- A behaviour-tree or state-machine diagram showing closed-loop system behaviour.
- A brief description of the function of each node.
- Any custom message types or interfaces should be listed and explained.

4. Technical Components

- **Computer Vision:** describe your vision pipeline and how it contributes to the task.
- **Custom End-Effector:** provide photos/renders, assembly details, engineering drawings, control overview and integration details.
- **System Visualisation:** explain how your system is visualised (e.g. RViz) and what it demonstrates.
- **Closed-Loop Operation:** describe the feedback method and how it adapts system behaviour in real time.

5. Installation and Setup

- Step-by-step installation instructions for dependencies and workspace setup.
- Hardware setup information (UR5e connection, camera, Teensy, etc.).
- Any environment variables, configuration files, or calibration procedures required to run the system. (You can take as assumed that there is some sort of hand-eye calibration already present in the system.)

6. Running the System

- Clear instructions for launching and running the complete system.
- Example commands (e.g. `ros2 launch project_name bringup.launch.py`).
- Expected behaviour and example outputs.
- Optional troubleshooting notes.
- The system should be launched by a single command (e.g. via a launch file, shell script or Docker image), without manual sequencing.

7. Results and Demonstration



- Describe how your system performs against its design goals.
- Include quantitative results where possible (e.g. accuracy, repeatability).
- Provide photos, figures, or videos showing the system in operation. (See Project Overview, above.)
- Highlight robustness, adaptability, and innovation.

8. Discussion and Future Work

- Briefly discuss major engineering challenges faced and how they were addressed.
- Outline opportunities for improvement or extensions – what would you do better for “Version 2.0”.
- Summarise what makes your approach novel, creative, or particularly effective.

9. Contributors and Roles

- Briefly list team members and describe their primary areas of responsibility (e.g. vision, planning, hardware).

10. Repository Structure

- A short section outlining the folder structure of your repository.
- Explain briefly what each main directory contains.

11. References and Acknowledgements:

- Credit any external libraries, tutorials, or prior codebases.
- Acknowledge external assistance (e.g. demonstrators, other groups).

3.1.2. Formatting and Submission

- The README must be named README.md and located in the root directory of your team’s GitHub repository.
- Ensure that all required images, diagrams, and media links render correctly on GitHub.
- All media (images, GIFs, diagrams) should be embedded or linked appropriately (no broken paths).
- Submission will be via a link to your repository on Moodle.

There are several ways to make your README publicly accessible, here are three suggestions:

1. Make your entire repository public
2. Fork/clone the repository, make it public and remove everything except the README
3. Use GitHub Pages or Gists to make a public-facing README

3.2. Peer Evaluation

Peer evaluation (sometimes called “Team Evaluation” on Moodle) will be used to scale the README marks to produce an individual mark. Peer evaluation for the README will be due at **17:00 on Friday of Week 12**.

- If you do not complete the peer assessment, you will be **penalised 10%** of the assignment mark (2% of the course mark). This is to ensure that all team members get constructive feedback.

There are three questions that make up the peer assessment:

1. **Relative quantity:** Each team member will rate the relative quantity of effort or contribution from each member of the team toward the preparation of the README.
2. **Relative quality:** Each team member will rate the relative quality of the work done by each member of the team.
3. **Comments:** There will be two comment fields:
 - i. A *compulsory* section asking for comments on your teammates – these will remain private.



- ii. An *optional* section asking for any public anonymous feedback you would like to give to your peers. Remember, you can provide positive as well as constructive feedback.



4. Marking Rubric

Competency	Developing	Competent	Proficient	Exceptional
Technical Documentation & Clarity	README unclear or incomplete; poor structure.	Clear in parts; minor confusion or missing details.	Well-structured, professional tone; logical flow.	Exceptionally clear, concise and professional; reads like industry documentation.
System Architecture & Technical Coverage	Missing or inaccurate node diagram; major gaps in component explanations.	Includes basic architecture with some explanation.	Complete architecture and technical coverage of all components.	Deep technical insight, excellent diagrams, shows mastery of ROS2 integration.
Reproducibility & Setup	Instructions missing or non-functional.	Basic setup steps provided.	Complete, usable installation/run guide.	Polished instructions with automation, troubleshooting, and clarity.
Results & Visual Presentation	No clear evidence of operation.	Minimal figures or unclear results.	Demonstrates system performance with clear visuals.	Highly professional presentation with clear video demonstration, visualisations and quantitative data.
Discussion & Future Work	Minimal commentary or vague ideas.	Some discussion of challenges and improvements.	Clear analysis of design trade-offs and next steps.	Insightful, well-reasoned engineering discussion; strong innovation awareness.